

PCA-ANN Face Recognition System Project Report

Introduction

This project implements a face recognition system using Principal Component Analysis (PCA) and machine learning classification techniques.

Objective

To develop a face recognition system using PCA (Eigenfaces) and evaluate classification performance with accuracy analysis and imposter detection.

Dataset

The dataset contains 9 enrolled individuals with 50 images each, resulting in 450 face images. Images were resized to 100x100 pixels and converted to grayscale.

Methodology

The workflow includes preprocessing, mean face computation, PCA-based feature extraction, Eigenface generation, classification, and evaluation.

PCA (Eigenfaces)

PCA reduces dimensionality while preserving important facial information. Eigenfaces are generated from the principal components.

Classification

Extracted features are used for classification. KNN was used for the final evaluation after testing multiple configurations.

Accuracy vs k Analysis

Different values of k were evaluated. Accuracy increased with larger k values, reaching a best accuracy of 57.78%.

Imposter Detection

Images not belonging to enrolled individuals were tested and successfully identified as 'Not Enrolled Person' using a distance threshold.

Results

Training Samples: 270, Testing Samples: 180, Best Accuracy: 57.78%. Confusion matrix and classification report were generated.

Conclusion

The project successfully demonstrates PCA-based face recognition with classification, evaluation, and imposter detection.